

ASSIGNMENT – III

Analysis of Assignment.pcap

1. What is the date of the transmission ?

Answer: To find the date, open the pcap file and check the frame timestamp.

Steps:

Open Assignment.pcap in Wireshark

Click on the first packet (No. 1)

Look at the "Frame" section in the middle pane

```
Frame 1: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits)
  Encapsulation type: Ethernet (1)
  Arrival Time: Sep  3, 2009 22:11:38.447250000 India Standard Time
  UTC Arrival Time: Sep  3, 2009 16:41:38.447250000 UTC
  Epoch Arrival Time: 1251996098.447250000
  [Time shift for this packet: 0.000000000 seconds]
```

2. What is the client's IP address ?

Answer: The client's IP address is the source IP of the initial packets.

Steps:

Open the pcap file

Look for the first packet (No. 1)

Check the "Internet Protocol Version 4" section

```
Frame 1: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits)
Ethernet II, Src: aa:aa:aa:bb:bb:bb (aa:aa:aa:bb:bb:bb), Dst: SenaoInterna_43:95:c2 (00:02:6f:43:95:c2)
Internet Protocol Version 4, Src: 10.100.1.24, Dst: 74.125.19.113
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
```

3. Destination Address

Answer: Multiple servers:

- **Primary (first handshake):** 74.125.19.113 (Google - clients1.google.com)
- **Secondary (second handshake):** 147.144.1.212 (www.ccsf.edu)

The capture shows two separate TCP connections to different destination servers.

4. How Many TCP Handshakes Are There?

Answer: 2 TCP handshakes

First Handshake (Google):

- Packets 1, 2, 3

Second Handshake (CCSF):

- Packets 10, 11, 12

Each handshake consists of:

- SYN (client → server)
- SYN-ACK (server → client)
- ACK (client → server)

5. Find the first handshake and write down the packet numbers of those packets (the column labeled "No.").

- **Filter:** `frame.number >= 1 && frame.number <= 3`

Answer: Packets 1, 2, and 3

Packet 1 (SYN):

- Source: 10.100.1.24, Port 51645
- Destination: 74.125.19.113, Port 80 (HTTP)
- Flags: S (SYN)
- Sequence Number: 3819318445

Packet 2 (SYN-ACK):

- Source: 74.125.19.113, Port 80
- Destination: 10.100.1.24, Port 51645
- Flags: SA (SYN-ACK)
- Sequence Number: 3879074656
- Acknowledgment: 3819318446

Packet 3 (ACK):

- Source: 10.100.1.24, Port 51645
- Destination: 74.125.19.113, Port 80
- Flags: A (ACK)
- Sequence Number: 3819318446
- Acknowledgment: 3879074657

6. How many HTTP GET request packets are there?

- Filter: `http.request.method == "GET"`

Answer: 3 HTTP GET request packets

1. **Packet 4** - GET /complete/search (Google Auto-Complete)
2. **Packet 13** - GET /Graphics/ccsfseal.gif (CCSF Seal Image)
3. **Packet 28** - GET /favicon.ico (CCSF Favicon)

7. Find the first HTTP GET request packet. What was the server's IP address?

- Filter: `frame.number == 4`

Answer: Packet 4

Server's IP Address: 74.125.19.113 (Google)

Request Details:

```
GET
/complete/search?client=chrome&hl=en-US&q=http%3A%2F%2Fwww.ccsf.edu%
2FGraphics%2Fccsfseal.gif HTTP/1.1
Host: clients1.google.com
Connection: keep-alive
User-Agent: Mozilla/5.0 (Windows; U; Windows NT 6.1; en-US)
AppleWebKit/530.5 (KHTML, like Gecko) Chrome/2.0.172.43 Safari/530.5
Accept-Encoding: gzip,deflate,sdch
Cookie:
PREF=ID=cbbab999d942c22a:TM=1250952115:LM=1250952162:GM=1:S=Ur1ivrKK
1EpeA_d5
Accept-Language: en-US,en;q=0.8
```

Accept-Charset: ISO-8859-1,utf-8;q=0.7,*;q=0.3

8. Illustrate the difference between the HTTP Packets numbered 4 and 6.

-Filter: `frame.number == 4 || frame.number == 6`

Packet 4: HTTP GET Request (Client → Server)

Type: HTTP Request

Direction: Outgoing (10.100.1.24 → 74.125.19.113)

TCP Flags: PA (Push-Acknowledge)

Payload Size: 494 bytes

Content:

- GET request for Google auto-complete search service
- Contains: URL, Host, Connection, User-Agent, Cookies, Accept headers

Key Characteristics:

- Initiates HTTP communication
- Contains request method (GET) and URI
- Includes client preferences (Accept-Encoding, Accept-Language)
- Sends cookies to server
- Sets Connection: keep-alive

Packet 6: HTTP Response (Server → Client)

Type: HTTP Response

Direction: Incoming (74.125.19.113 → 10.100.1.24)

TCP Flags: PA (Push-Acknowledge)

Payload Size: 347 bytes

Content:

HTTP/1.1 200 OK

Content-Type: text/javascript; charset=utf-8

Date: Thu, 03 Sep 2009 16:41:26 GMT

Expires: Thu, 03 Sep 2009 17:41:26 GMT

Cache-Control: public, max-age=3600

Content-Encoding: gzip

```
Server: Auto-Completion Server
Content-Length: 92
```

Key Characteristics:

- Server response to request
- Contains HTTP status line (200 OK)
- Includes response headers (Content-Type, Cache-Control, etc.)
- Body is gzip-compressed JavaScript data (92 bytes)
- Sets cache expiration (1 hour)

9. Give a complete analysis of the packet numbered 20 ?

Layer 2 - Ethernet Frame

```
Destination MAC: aa:aa:aa:bb:bb:bb (Client)
Source MAC: 00:02:6f:43:95:c2 (Server/Gateway)
Type: IPv4 (0x0800)
```

Layer 3 - IP Header

```
Version: 4
Header Length (IHL): 5 (20 bytes)
Type of Service (ToS): 0x0 (no special handling)
Total Length: 1448 bytes (1420 data + 28 header)
Identification: 40043
Flags: DF (Don't Fragment)
Fragment Offset: 0
TTL (Time To Live): 50
Protocol: TCP (6)
Header Checksum: 0x605
Source IP: 147.144.1.212 (CCSF Server)
Destination IP: 10.100.1.24 (Client)
```

Layer 4 - TCP Header

```
Source Port: 80 (HTTP)
```

```
Destination Port: 51646 (Client ephemeral port)
Sequence Number: 2769882623
Acknowledgment Number: 1583945190
Data Offset: 5 (20 bytes header, no options)
Reserved: 0
Flags: A (ACK only - acknowledgment flag)
Window Size: 33088 bytes
Checksum: 0x917e
Urgent Pointer: 0
Options: None
```

Layer 5+ - Application Data (HTTP Response Payload)

```
Total Payload: 1408 bytes (1448 - 20 IP header - 20 TCP header)
Content: Partial GIF image data (continuation)
```

Purpose

Packet 20 is the 4th data segment of the HTTP response body containing portions of the GIF image file. It's a continuation of the data stream started in Packet 14. The ACK flag indicates it's also acknowledging receipt of data from the client.

10. Is the connection persistent? If yes, how is it recognized?

Look at packet PORT NUMBERS:

- **Packet 13** (GET /Graphics/ccsfseal.gif): Port **51646** ✓
- **Packet 28** (GET /favicon.ico): Port **51646** ✓

Same port = Same connection = **PERSISTENT** ✓

If the connection closed, packet 28 would use a different port (like 51647). But it uses the **same port 51646**, proving the connection stayed open.